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Rotating joint for crossbar leveling adjustment

Abstract

A monitor mounting system which includes a crossbar having at least one monitor mounting bracket positioned thereon. A support arm is configured to adjust the vertical height of the crossbar and a pivot assembly is positioned between the support arm and the crossbar, with the pivot assembly configured to allow limited pivot adjustment of the crossbar in order to level the crossbar to the horizontal position.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS/PRIORITY (1) This application is a continuation-in-part of U.S. Ser. No. 17/876,912 filed on Jul. 29, 2022, and which is hereby incorporated by reference in its entirety.

BACKGROUND
(1) With computer monitors become increasingly ubiquitous at work and home, individuals are spending ever greater portions of their waking hours viewing monitors. This means highly customizable adjustment of a monitor's position is necessary to reduce the strain of viewing

monitors for long periods. FIG. **1** illustrates a convention monitor stand **100** supporting a monitor **101**. Monitor stand **100** includes base **102**, upright **103**, and monitor mounting bracket **104**. FIG. **1** suggests how monitor stand **100** allows monitor **101** to be adjusted with four degrees of freedom: (1) height in a vertical direction along upright **103**, (2) swivel of upright **103** with respect to base **102**, (3) tilt of monitor top and bottom edge toward and away from the user; and (4) pivot of the monitor in a plane containing the monitor. “Vertical” or “vertical direction” as used herein means a direction parallel to Earth's gravitational force. While a single monitor stand such as seen in FIG. **1** can adequately provide these four degrees of freedom, providing this degree of adjustability becomes more difficult as stands are expected to support two, three, or possibly more monitors.

SUMMARY

(2) One embodiment of the invention is a monitor mounting system which includes a crossbar having at least one monitor mounting bracket positioned thereon. A support arm is configured to adjust the vertical height of the crossbar and a pivot assembly is positioned between the support arm and the crossbar, with the pivot assembly configured to allow limited pivot adjustment of the crossbar in order to level the crossbar to the horizontal position.

(3) The above summary is not intended to describe each illustrated embodiment or every possible implementation. These and other features, aspects, and advantages of the present invention will become better understood with regard to the following description, appended claims, and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying figures, where like reference numerals refer to identical or functionally similar elements throughout the separate views, which are not true to scale, and which, together with the detailed description below, are incorporated in and form part of the specification, serve to illustrate further various embodiments and to explain various principles and advantages in accordance with the present invention:

(2) FIG. **1** is a perspective view of a prior art monitor stand.

(3) FIG. **2** is a perspective view of one embodiment of the present invention.

(4) FIG. **3**A is a perspective view of one embodiment of the pivot assembly.

(5) FIG. **3**B is a cross-sectional view of the pivot assembly of FIG. **3**A.

(6) FIG. **4**A is an exploded view of the pivot assembly from a first direction.

(7) FIG. **4**B is an exploded view of the pivot assembly from an opposing direction.

(8) FIG. **5**A is a vertical cross-sectional view of another embodiment of the pivot assembly.

(9) FIG. **5**B is a horizontal cross-sectional view of the pivot assembly depicted in FIG. **5**A.

(10) FIG. **6**A is a front exploded view of the pivot assembly depicted in FIG. **5**A.

(11) FIG. **6**B is a rear exploded view of the pivot assembly depicted in FIG. **5**A.

(12) FIG. **7** is a perspective view of the inner face of the pivot core of the pivot assembly depicted in FIG. **5**A.

DETAILED DESCRIPTION

(13) Detailed embodiments of the present invention are disclosed herein; however, it is to be understood that the disclosed embodiments are merely exemplary of the invention, which can be embodied in various forms. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a basis for the claims and as a representative basis for teaching one skilled in the art to variously employ the present invention in virtually any appropriately detailed structure. Alternate embodiments may be devised without departing from the spirit or the scope of the invention. Further, the terms and phrases used herein are not intended to be limiting; but rather, to provide an understandable description of the invention. While the

specification concludes with claims defining the features of the invention that are regarded as novel, it is believed that the invention will be better understood from a consideration of the following description in conjunction with the drawing figures, in which like reference numerals are carried forward.
(14) As used herein, the terms “a” or “an” are defined as one or more than one. The term “plurality,” as used herein, is defined as two or more than two. The term “another,” as used herein, is defined as at least a second or more. The terms “comprises,” “comprising,” or any other variation thereof are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements, but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element proceeded by “comprises . . . a” does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises the element. The terms “including,” “having,” or “featuring,” as used herein, are defined as comprising (i.e., open language). The term “coupled,” as used herein, is defined as connected, although not necessarily directly, and not necessarily mechanically. As used herein, the term “about” or “approximately” applies to all numeric values, whether or not explicitly indicated. These terms generally refer to a range of numbers that one of skill in the art would consider equivalent to the recited values (i.e., having the same function or result). In many instances these terms may include numbers that are rounded to the nearest significant figure. Where a numerical limitation is used, unless indicated otherwise by the context, “about” or “approximately” means the numerical value can vary by +/−5%, +/−10%, or in certain embodiments +/−15%, or possibly as much as +/−20%. Similarly, the term “substantially” will typically mean at least 85% to 99% of the characteristic modified by the term. For example, “substantially all” will mean at least 85%, at least 90%, or at least 95%, etc. Relational terms such as first and second, top and bottom, right and left, and the like may be used solely to distinguish one entity or action from another entity or action without necessarily requiring or implying any actual such relationship or order between such entities or actions.
(15) Described now are exemplary embodiments of the present invention. FIG. **2** illustrates one embodiment of the present invention, tilt adjustable monitor crossbar assembly **1**, sometimes referred to simply as monitor mounting system **1**. In this embodiment, the monitor mounting assembly **1** most generally includes a crossbar **5**, a support arm **15**, and a pivot assembly or pivot link **25** connected between crossbar **5** and support arm **15**. In the FIG. **2** embodiment, crossbar **5** includes three crossbar segments **6** with hinges **7** connecting the interior ends of the crossbar segments **6** to each other, allowing the outer crossbar segments to swing inward toward the center crossbar segment. Typically, each crossbar segment **6** will have a monitor mounting bracket **10** attached thereto. In FIG. **2**, the monitor mounting brackets include a mounting face **11** for attachment to the monitor and a hinged connection or tilt connector **12** allowing for altering of the tilt position of the monitor with respect to its crossbar segment **6**. Although FIG. **2** illustrates a crossbar **5** formed of multiple hinged segments, other embodiments could employ a single segment crossbar.
(16) FIG. **2** further shows the support arm **15** which is connected to the support surface **90** (e.g., a work desk) and suspends crossbar **5** via pivot assembly **25** above support surface **90**. Although support arm **15** could be almost any conventional or future developed support mechanism, the FIG. **2** embodiment illustrates a support arm available from Humanscale Corporation under the tradenames M8.1 and M10. Although not shown in FIG. **2**, another arm may be positioned between the support surface and the base swivel connection **16** of support arm **15**. The base swivel connection **16** will allow the support arm to rotate in the swivel direction and also an upper swivel fork **17** will operate in conjunction with the pivot assembly **25** to create a second swivel point. In the example of M8.1 or M10 support arms, the support arms include a weight compensating spring system which allows for change of the vertical position of support arm **15** (i.e., change of the

vertical height of crossbar **5**) while the user experiences little or no change in the force needed to adjust the height of the monitors mounted on crossbar **5**. The construction and operation of the illustrated support arm **15** is more fully described in U.S. Pat. No. 10,480,709 which is incorporated by reference herein.

(17) As suggested above, pivot assembly **25** provides the connection between crossbar **5** and support arm **15**. FIGS. **3**A to **4**B illustrate one embodiment of pivot assembly **25**. This embodiment of pivot assembly **25** will generally include pivot housing **26** which is integrally formed on one end with swivel ring **50** and is engaged on the opposite end by pivot core **35**. Swivel ring **50** will include two bearing inserts **51** (best seen in FIG. **4**B). Bearing inserts **51** may be formed of a low friction material such as polyoxymethylene (POM), also known as acetal, polyacetal, or polyformaldehyde. A pin (not shown) with an alignment key will extend through the upper swivel fork **17** of support arm **15** (see FIG. **2**) and complete the swivel connection between swivel fork **17** and swivel ring **50**. The bearing flexures **52** allow for a controlled precision fit between a rotation pin (not shown) and the contact surface of bearing inserts **51**.

(18) As best seen in FIG. **4**B, pivot housing **26** is mainly a hollow body with internal threads **27** formed on the inner surface of pivot housing **26**. A threaded set screw aperture **29** extends through the housing wall to accommodate set screw **30** which has sufficient length to extend into internal rotation surface of pivot core **35**. The face of pivot housing **26** to which swivel ring **50** attaches (see FIG. **3**B) includes the limit screw apertures **28** extending through this face, into the interior of pivot housing **26**, and then into a controlled rotary slots (or "pivot slots") **37** in pivot core **35**.

(19) The internal threads **27** of pivot housing **26** will be engaged by the external threads **36** of pivot core **35** with set screw **30** able to engage external threads **36** in order to selectively lock the relative rotative positions between pivot housing **26** and pivot core **35**. As seen in FIG. **3**B, pivot core **35** includes the controlled rotary slots **37** formed on the inner face of pivot core **35**. Pivot core **35** also includes friction pin aperture **38** extending through pivot core **35** and a series of crossbar mounting lug apertures **39** (better seen in FIG. **4**B) formed in the external face of pivot core **35**. As best seen viewing FIGS. **3**B and **4**A, the motion control plate (sometimes called an "anti-rotation plate") **40** and the compression plate **43** are positioned between the inner faces of pivot housing **26** and pivot core **35**. FIG. **4**A shows how the elliptically shaped motion control plate **40** will include opposing circumferential cutouts **41** and a plate groove or plate inset **44**, which does not extend completely through motion control plate **40** and is shaped to accommodate compression plate **43**. FIG. **3**B best shows how motion control plate **40** and compression plate **43** are sandwiched between the inner faces of pivot housing **26** and pivot core **35**. The plate inset **44** will ensure there is no relative movement between motion control plate **40** and compression plate **43**. In one preferred embodiment, motion control plate **40** is formed of polyurethane foam and compression plate **43** is formed of acrylonitrile butadiene styrene. Angle limit screws **47** will extend through pivot housing **26**, the circumferential cutouts **41** in motion control plate **40**, and into the controlled rotary slots **37** of pivot core **35**.

(20) Viewing FIG. **4**A, it can be envisioned how angle limit screws **47** will allow a limited relative rotation between pivot housing **26** and pivot core **35** based on the angular width of controlled rotary slots **37** and circumferential cutouts **41** (at least when set screw **30** in not engaging external threads **36** of pivot core **35**). The resulting degree of pivot imparted to crossbar **5** is suggested by the angle θ seen in FIG. **2**. The angle θ represents the degree of tilt (in one direction) of crossbar **5** relative to the swivel ring **50** fixed against tilt by upper swivel fork **17** of support arm **15**. The maximum pivot of crossbar **5** (pivot in both directions) would be 20. In certain embodiments, the width of controlled rotary slots **37** and circumferential cutouts **41** are only wide enough to allow a maximum pivot of less than 15 degrees (θ=7.5 degrees or less), and more preferably a maximum pivot of 8 degrees (θ=4 degrees or less), between pivot housing **26** and pivot core **35**. Returning to FIG. **3**B, it may also be envisioned how the compression screw **45** engages motion control plate **40** and will tend to compress compression plate **43** against the inner face of pivot housing **26**. It will

be understood that advancing compression screw **45** into motion control plate **40** will increase the amount of torque necessary to cause rotation between pivot housing **26** and pivot core **35**. In addition to enhancing friction within the system, motion control plate **40** provides a locating feature for assembly of the system.

(21) Viewing FIG. **2**, with compression screw **45** applying moderate force to compression plate **43**, a user will be allowed to finely adjust the pivot position of crossbar **5** by applying moderate force to the outer monitors and overcoming the friction generated by compression plate **43** and motion control plate **40**. However, when the user ceases applying force to the monitors, the friction is sufficient to resist the weight of the monitors correcting the user induced pivot. Thereafter, set screw **30** may be tightened to engage external threads **36** on pivot core **35** in order to more firmly lock the pivot position of crossbar **5**. The use of compression screw **45** may be particularly useful when heavier monitors are being mounted, but there could be alternative embodiments which do not employ a compression screw **45**. Similarly, while many embodiments employ motion control plate **40** and compression plate **43** to enhance the friction resisting free pivot motion of the crossbar, there could be alternative embodiments which eliminate motion control plate **40** and compression plate **43**.

(22) Another embodiment of the pivot assembly **25** is depicted in FIGS. **5**A-**7**. Turning to FIG. **5**A, a vertical cross-section—comparable to FIG. **3**A—of this embodiment is shown. This embodiment of pivot assembly **25**, like the embodiment described above, will generally include pivot housing **210** which is integrally formed on one end with swivel ring **222** and is engaged on the opposite end by pivot core **230**. Pivot housing **210** is mainly a hollow body with internal threads **212** formed on its inner surface that engage with external threads **232** formed on the outer surface of pivot core **230**. The face of pivot housing **210** to which swivel ring **222** attaches includes limit screw apertures **214** extending into the interior of pivot housing **210**. When pivot core **230** is threaded into pivot housing **210**, the limit screw apertures **214** in the pivot housing **210** align with controlled rotary slots **234** in the internal face of pivot core **230**. Although the angle limit screws **216** are firmly secured to the pivot housing **210** by the limit screw apertures **214**, the elliptical shape of the controlled rotary slots **234** (see FIG. **7**) allows the angle limit screws **216** to slide back and forth within the slots **234**. This sliding engagement between angle limit screws **216** and controlled rotary slots **234** allows limited relative rotation between pivot housing **210** and pivot core **230**. This rotational motion is facilitated by the mating of the internal threads of the pivot housing **210** with the external threads of the pivot core **230**. Referring now to FIG. **5**B, crossbar **5** mounts to pivot core **230** via threaded engagement between mounting screws (not pictured) and a series of threaded crossbar mounting apertures **236** that extend from the pivot core's **230** external face through to its internal face (see the ingress and egress of crossbar mounting apertures **236** in FIGS. **6**A-**7**). Thus, when pivot core **230** rotates relative to the pivot housing **210**, so does crossbar **5**.

(23) As best seen when viewing FIGS. **5**A and **5**B, this embodiment differs from the embodiment described above because it utilizes a motion control pad **238** to provide the resistive force rather than motion control plate **40** and compression plate **43**. Motion control pad **238** is compressed between the inner faces of pivot housing **210** and pivot core **230**, providing resistance to prevent the pivot core **230** from sliding freely around angle limit screws **216** and thereby securing the crossbar **5** in the user's desired position. In one preferred embodiment, motion control pad **238** is formed of Cellasto® microcellular polyurethane elastomer, but in other embodiments motion control pad could be formed of any shape-memory material (i.e., tends to return to its original size and shape) that is highly compressive and durable in rotation and compression. The internal face of pivot housing **210** comprises detents **218** (shown in FIG. **5**B) into which motion control pad **238** is compressed when pivot assembly **25** is assembled, and angle limit screws **216** (shown in FIG. **5**A) are threaded through motion control pad **238** before entering controlled rotary slots **234**. The detents **218** and angle limit screws **216** firmly secure the motion control pad **238** to the pivot housing **210**. As pivot core **230** is threadingly rotated into pivot housing **210**, it compresses motion

control pad **238**, reducing the thickness of the pad **238** and causing it to deform into the pivot core **230**, thereby creating a resistive force that prevents the pivot core **230** from rotating relative to the pivot housing **210**. Within the monitor mounting system of FIG. **2**, the compression pad **238** allows a user to finely adjust the pivot position of crossbar **5** by applying moderate force to the outer monitors and overcoming the resistive force generated by compression pad **238**. However, when the user ceases applying force to the monitors, the resistive force is sufficient to resist the weight of the monitors correcting the user-induced pivot. The Cellasto® material of the motion control pad **238** enhances the performance of the leveling function by increasing the damping and friction force as compared to prior art designs. With this stronger resistive force, the compression pad **238** can maintain the crossbar's position without additional screws or other fastening means. Thus, this embodiment eliminates friction pin aperture **38** with compression screw **45** and threaded screw aperture **29** with set screw **30**, although these features may be combined with motion control pad **238** in other embodiments. This simplification of the design creates a more seamless experience with the user as well as reducing cost and environmental waste.
(24) Turning to FIGS. **6**A-**6**B, another modification in this embodiment of pivot assembly **25** is seen in bearing inserts **224**, **226**. Swivel ring **222** will include two bearing inserts **224**, **226** (best seen in), which create a smooth surface for rotation, reduce friction, and eliminate audible noise when rotating. First bearing insert **224** has a larger diameter than second bearing insert **226** in order to maximize stress dispersion and enhance rotational capability. The first bearing insert **224** is compressed to the inner surface of the swivel ring **222**, providing support in rotation and preventing angular pivot due to the cantilevered load. Bearing inserts **224**, **226** may be formed of a low friction material such as polyoxymethylene (POM), also known as acetal, polyacetal, or polyformaldehyde. The bearing inserts **224**, **226** comprise bearing flexures **228** that allow for a controlled precision fit between a rotation pin (not shown) and the contact surface of bearing inserts **224**, **226**.
(25) The foregoing description and accompanying drawings illustrate the principles, exemplary embodiments, and modes of operation of the invention. However, the invention should not be construed as being limited to the particular embodiments discussed above. Many modifications of the embodiments described herein will come to mind to one skilled in the art having the benefit of the teaching presented in the foregoing descriptions and the associated drawings. Accordingly, it should be appreciated that variations to those embodiments can be made by those skilled in the art without departing from the scope of the invention.

Claims

1. A monitor mounting system comprising: a) a crossbar having at least one monitor mounting bracket positioned thereon; b) a support arm configured to adjust the vertical height of the crossbar; c) a pivot assembly positioned between the support arm and the crossbar, wherein the pivot assembly comprises: (i) a pivot assembly housing comprising: a pivot core threaded into the pivot housing, wherein an inner face of the pivot core includes opposing circumferential slots; (ii) a swivel ring attached to an exterior of the pivot assembly housing; and (iii) a pivot stop positioned in the pivot assembly housing and limiting pivot adjustment to fifteen or fewer degrees in each pivot direction, wherein the pivot stop includes two angle limit screws engaging the circumferential slots of the pivot core and a compression pad situated inside the pivot housing.
2. The monitor mounting system of claim 1, wherein the two angle limit screws are threaded through apertures in the pivot housing, then through the compression pad, and finally extend into the circumferential slots of the pivot core.
3. The monitor mounting system of claim 1, wherein an outer face of the pivot core includes at least two threaded apertures configured to receive mounting screws.
4. The monitor mounting system of claim 1, wherein the swivel ring further includes two bearing

inserts.
5. The monitor mounting system of claim 1, wherein the crossbar has at least two monitor mounting brackets positioned thereon, and the monitor mounting brackets are attached to the crossbar with a tilt connector.
6. The monitor mounting system of claim 5, wherein the crossbar includes a central segment and two outer segments which form hinge connections to the central segment.
7. The monitor mounting system of claim 1, wherein the pivot assembly housing includes detents in an inner face of the pivot assembly housing, wherein the compression pad is compressed into the detents when the pivot core is threaded into the pivot assembly housing.
8. A monitor mounting system comprising: a. a crossbar having at least one monitor mounting bracket positioned thereon; b. a support arm configured to adjust the vertical height of the crossbar; c. a pivot assembly positioned between the support arm and the crossbar and configured to allow limited pivot adjustment of the crossbar, wherein the pivot assembly comprises: i. a pivot core, wherein an outer face of the pivot core includes at least two threaded apertures configured to receive mounting screws and wherein an inner face of the pivot core includes opposing circumferential slots; ii. a pivot assembly housing, wherein the pivot core is threaded into the pivot housing; and iii. a pivot stop positioned in the pivot housing and limiting pivot adjustment to fifteen or fewer degrees in each pivot direction.
9. The monitor mounting system of claim 8, wherein the circumferential slots have an arc of less than fifteen degrees.
10. The monitor mounting system of claim 8, wherein the pivot stop includes two angle limit screws engaging the pivot core and a compression pad situated in the pivot housing.
11. The monitor mounting system of claim 10, wherein the two angle limit screws are threaded through apertures in the pivot housing, then through the compression pad, and finally extend into the circumferential slots of the pivot core.
12. The monitor mounting system of claim 8, wherein the crossbar has at least two monitor mounting brackets positioned thereon, and the monitor mounting brackets are attached to the crossbar with a tilt connector.
13. The monitor mounting system of claim 8, wherein the crossbar includes a central segment and two outer segments which form hinge connections to the central segment.
14. The monitor mounting system of claim 10, wherein the pivot assembly housing includes detents in an inner face of the pivot assembly housing, wherein the compression pad is compressed into the detents when the pivot core is threaded into the pivot assembly housing.